Annesha Chakraborty

the.sea.flies29@gmail.com | +91 9123383321 | GitHub | LinkedIn

EDUCATION

Indian Institute of Technology Madras **2025 – Present**
BS in Data Science and Applications *Expected 2029*

TECHNICAL SKILLS

Programming & Data: Python, SQL, Pandas, NumPy, MySQL
Machine Learning: Scikit-learn, XGBoost, Classification, Regression, Feature Engineering, Model Validation, Cross-Validation, Time-Series Forecasting
Deep Learning: TensorFlow, Keras, ANN, CNN, Image Classification
Data Processing & APIs: Data Preprocessing, Data Cleaning, Data Transformation, REST APIs, FastAPI, Flask
Cloud & Deployment: Google Cloud Run, Docker, Git, GitHub
Core: Probability, Statistics, Linear Algebra, DBMS, Database Design, Normalization

PROJECTS

AgroVani – "Uber for Farmers" | Residue & Stubble Management System GitHub

- **Team Lead** for an agricultural AI platform focused on crop-residue management, monetization, and farmer decision support; won **HACK CORE 2026** at IIT Ropar.
- Integrated **Syngenta and CEHub agricultural APIs**, validating and transforming weather and crop data to generate **spray-window and abiotic-stress indicators**.
- Built **Crop Cam**, a plant-disease detection pipeline with ML inference API and recommendation engine for actionable crop-treatment guidance.
- Integrated the **Gemini API** for Voice Advisory and conversational agricultural guidance; contributed to AI-driven residue planning and farmer–buyer workflows.

Aditya-L1 Solar Flare Nowcasting & 30-Min Forecasting GitHub

- Processed and harmonized multi-sensor **SoLEXS and HEL1OS X-ray telemetry**, engineering temporal, rolling-statistical, lag, volatility, flux-ratio, and cumulative-energy features.
- Built **XGBoost classification and regression models** with time-aware **TimeSeriesSplit** validation for solar-flare detection and 30-minute forecasting.
- Achieved **97% TPR and 98% precision** for flare detection; regression achieved **30.34 MAE and 67.61 RMSE** on the evaluated dataset.

PlantMD – Plant Disease Detection API GitHub

- Developed a **TensorFlow/Keras deep-learning inference pipeline** for plant-disease image classification across **38 disease classes**, including image preprocessing and confidence scoring.
- Built a **FastAPI REST service** for image-based model inference and structured prediction responses; implemented validation and disease/symptom data APIs.
- Containerized the inference service with **Docker** and deployed the ML API on **Google Cloud Run** for cloud-based inference.

Bean-Matched Coffee Roast Predictor GitHub

- Built a multi-class ML pipeline using **Python, Pandas, Scikit-learn, and XGBoost**, including preprocessing, feature engineering, model comparison, and evaluation.
- Engineered interaction features from coffee preferences and behavior; addressed **class imbalance** using class-weighted XGBoost and evaluated performance using macro-F1 and minority-class recall.
- Improved minority-class recall from **0.12 to 0.45** through model/feature adjustments while analyzing the trade-off between overall accuracy and minority-class performance.

LEADERSHIP & EXPERIENCE

International Coding Mentor
Programming & Technical Education

- Mentored **100+ students 1:1 internationally** through programming guidance, debugging, project reviews, and technical roadmaps; designed and delivered a **10-day Pygame bootcamp (USA)**. Collaborated with a **startup (Coding Live)** to deliver an end to end ML course.

ACHIEVEMENTS

- **Deloitte Australia – Data Analytics Job Simulation** – Completed telemetry data transformation, dataset cleaning, and interactive **Tableau dashboard** development. [Certificate Link]
- **Zonal Round Selected Participant** – Women Who Master Hackathon 2026 [Aspire for Her × Logitech]. [Certificate Link]
- **Hackcore 2026 Winner** – ANNAM.ai × Syngenta Biologicals (IIT Ropar).